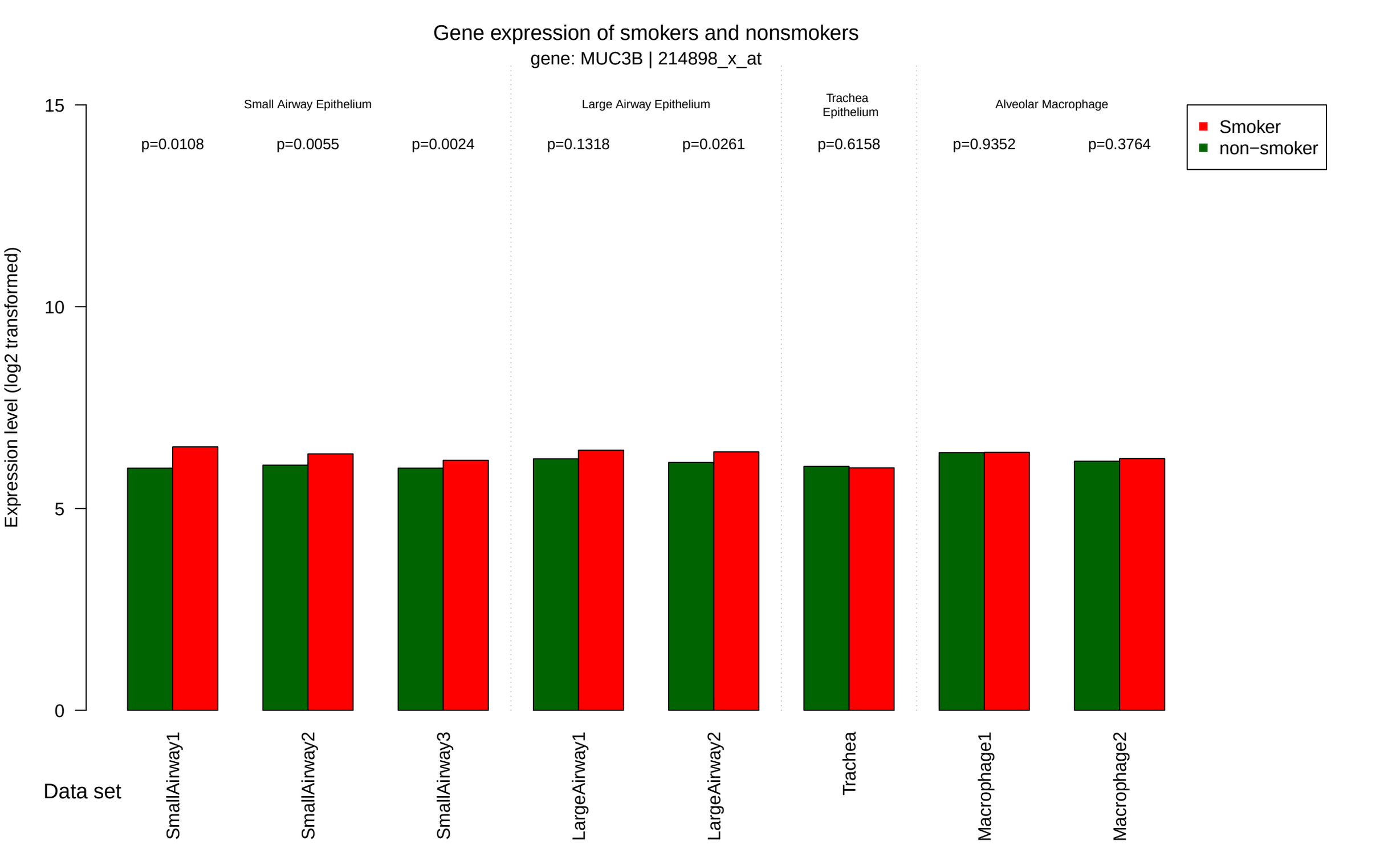
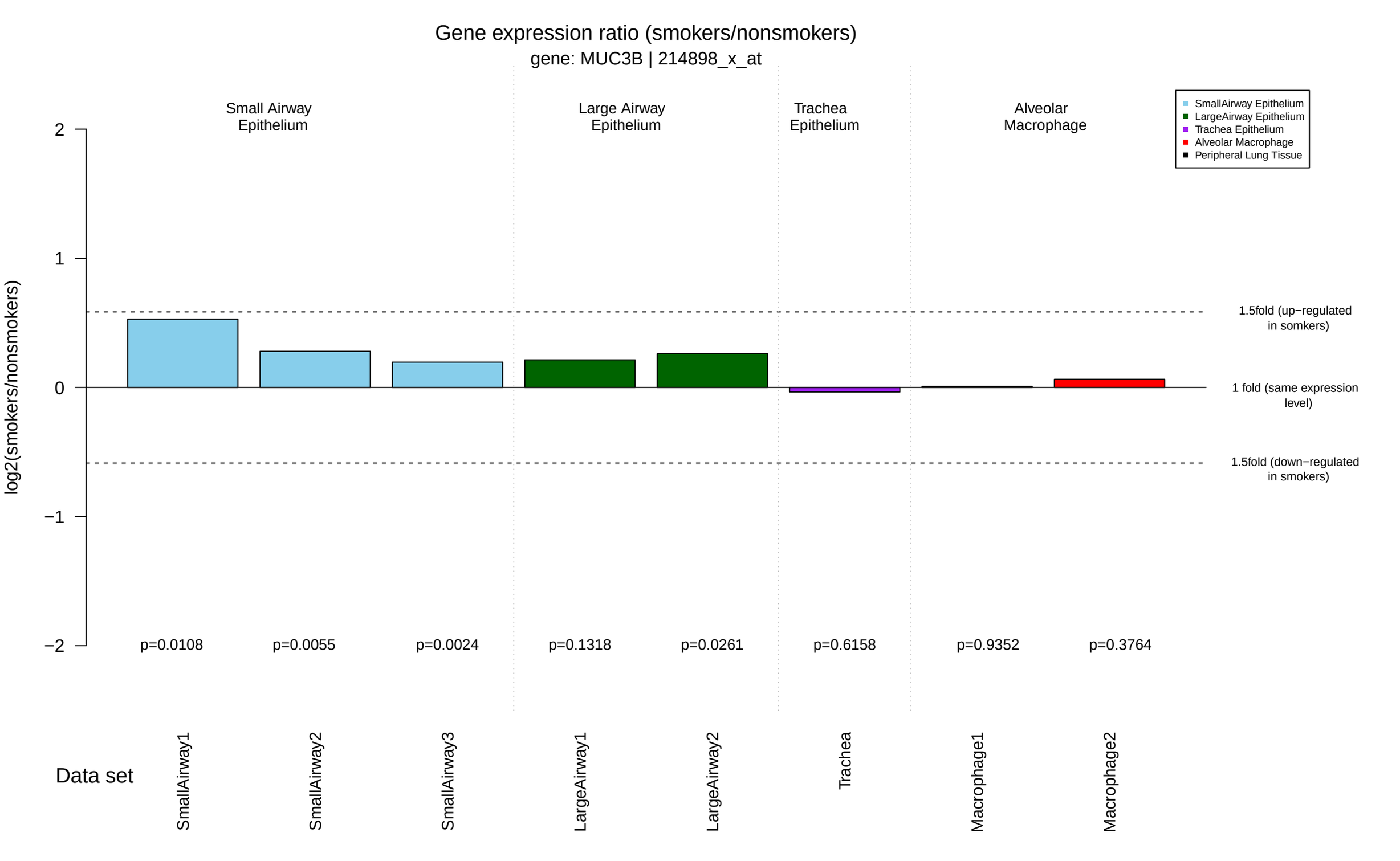
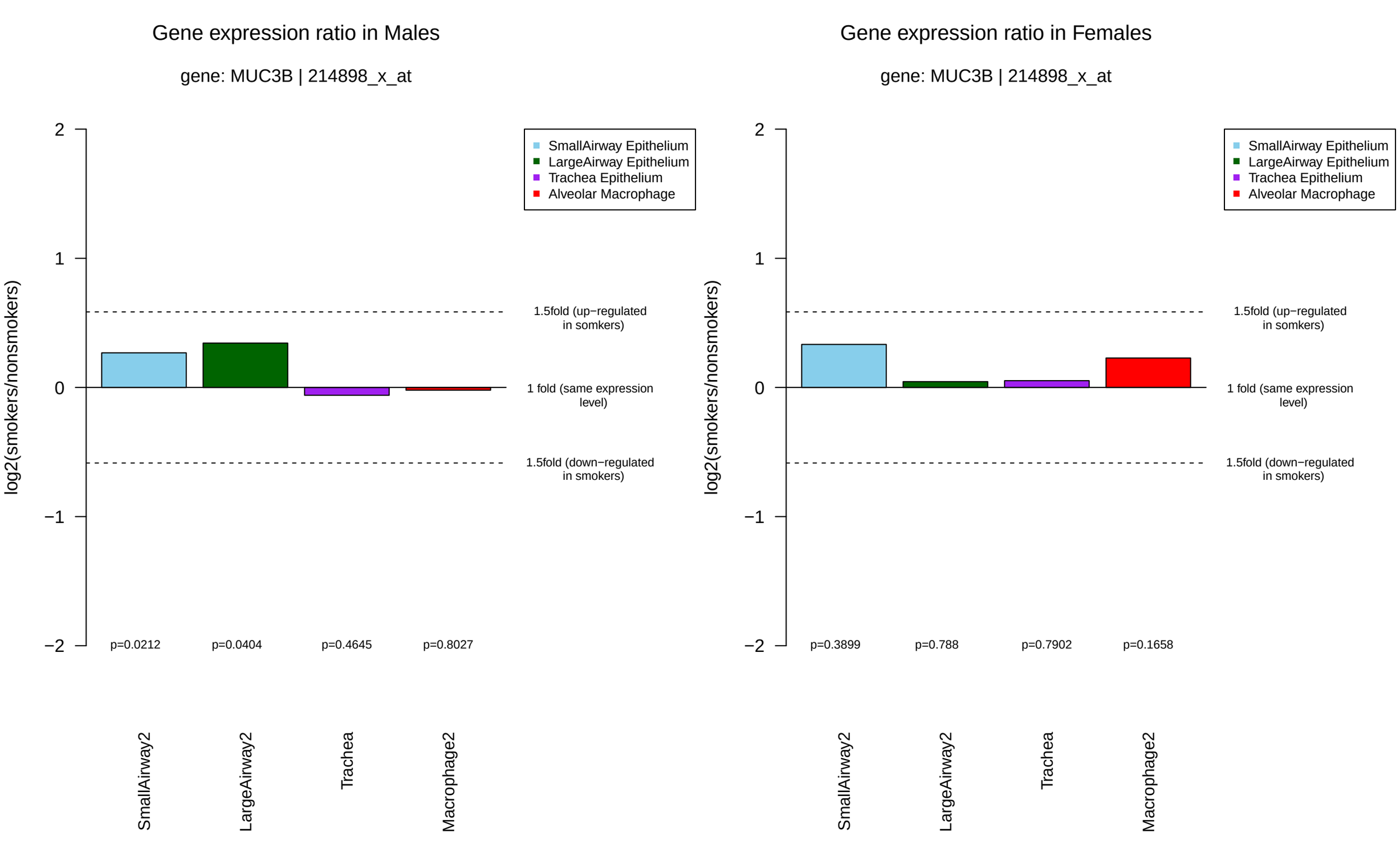
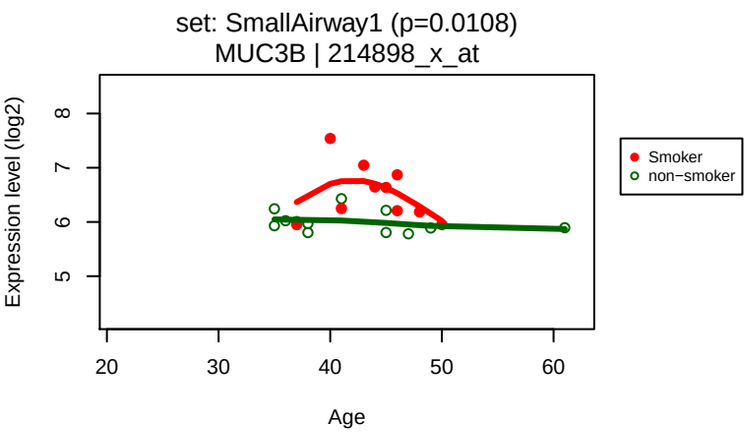
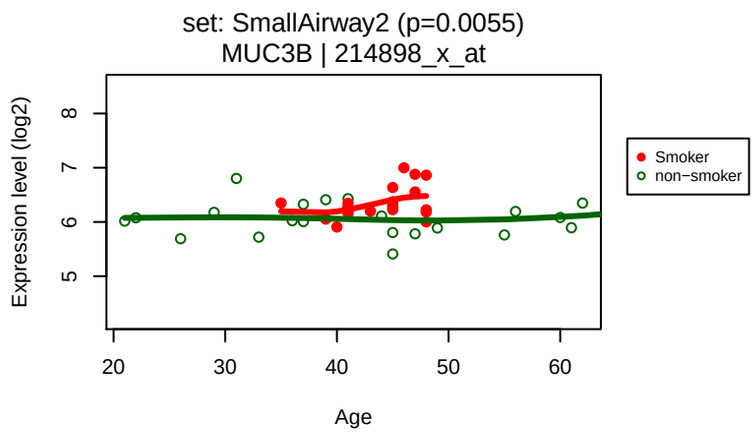
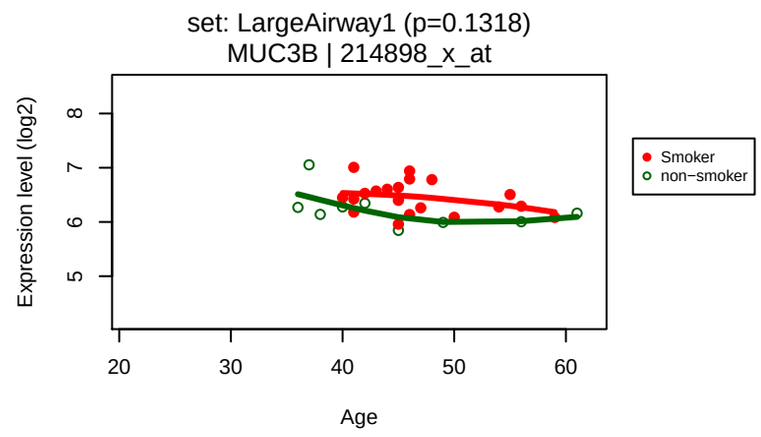

Gene expression of smokers and nonsmokers
gene: MUC3B | 214898_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0108
p=0.0055
p=0.0024
p=0.1318
p=0.0261
p=0.6158
p=0.9352
p=0.3764
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

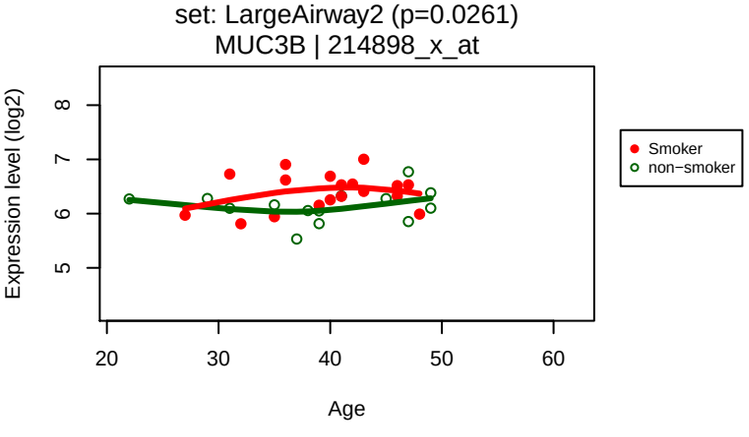

Gene expression ratio (smokers/nonsmokers)
gene: MUC3B | 214898_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0108
p=0.0055
p=0.0024
p=0.1318
p=0.0261
p=0.6158
p=0.9352
p=0.3764
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

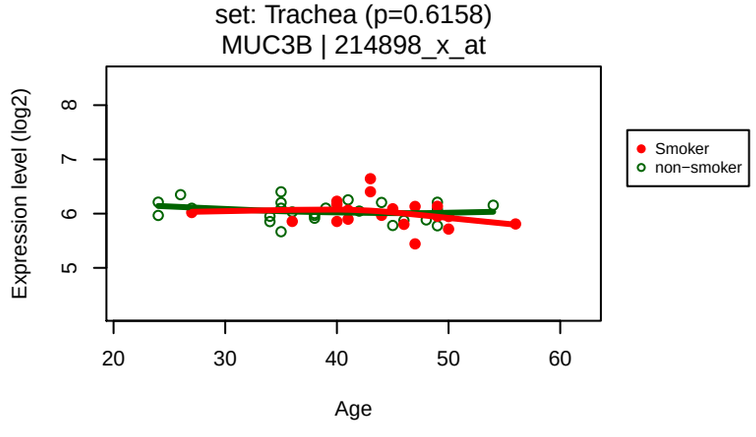

Gene expression ratio in Males
gene: MUC3B | 214898_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0212
p=0.0404
p=0.4645
p=0.8027
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MUC3B | 214898_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3899
p=0.788
p=0.7902
p=0.1658
SmallAirway2
LargeAirway2
Trachea
Macrophage2

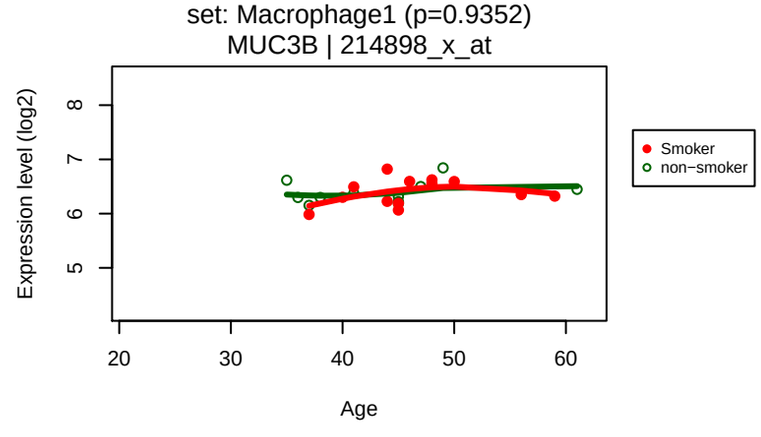

set: SmallAirway1 (p=0.0108)
MUC3B | 214898_x_at
Expression level (log2)
Age
Smoker
non−smoker

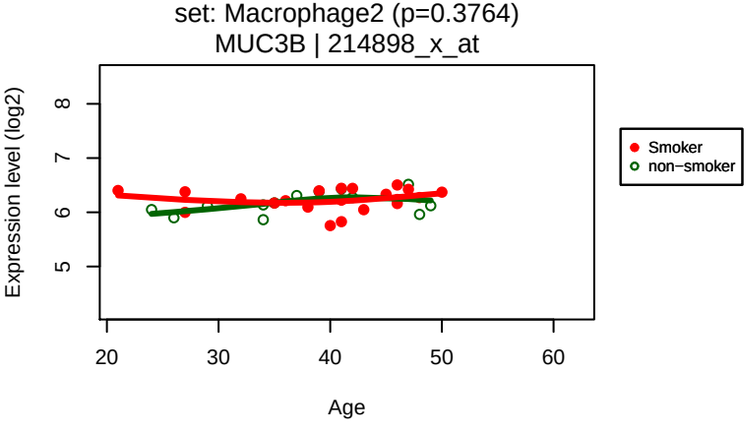

set: SmallAirway2 (p=0.0055)
MUC3B | 214898_x_at
Expression level (log2)
Age
Smoker
non−smoker

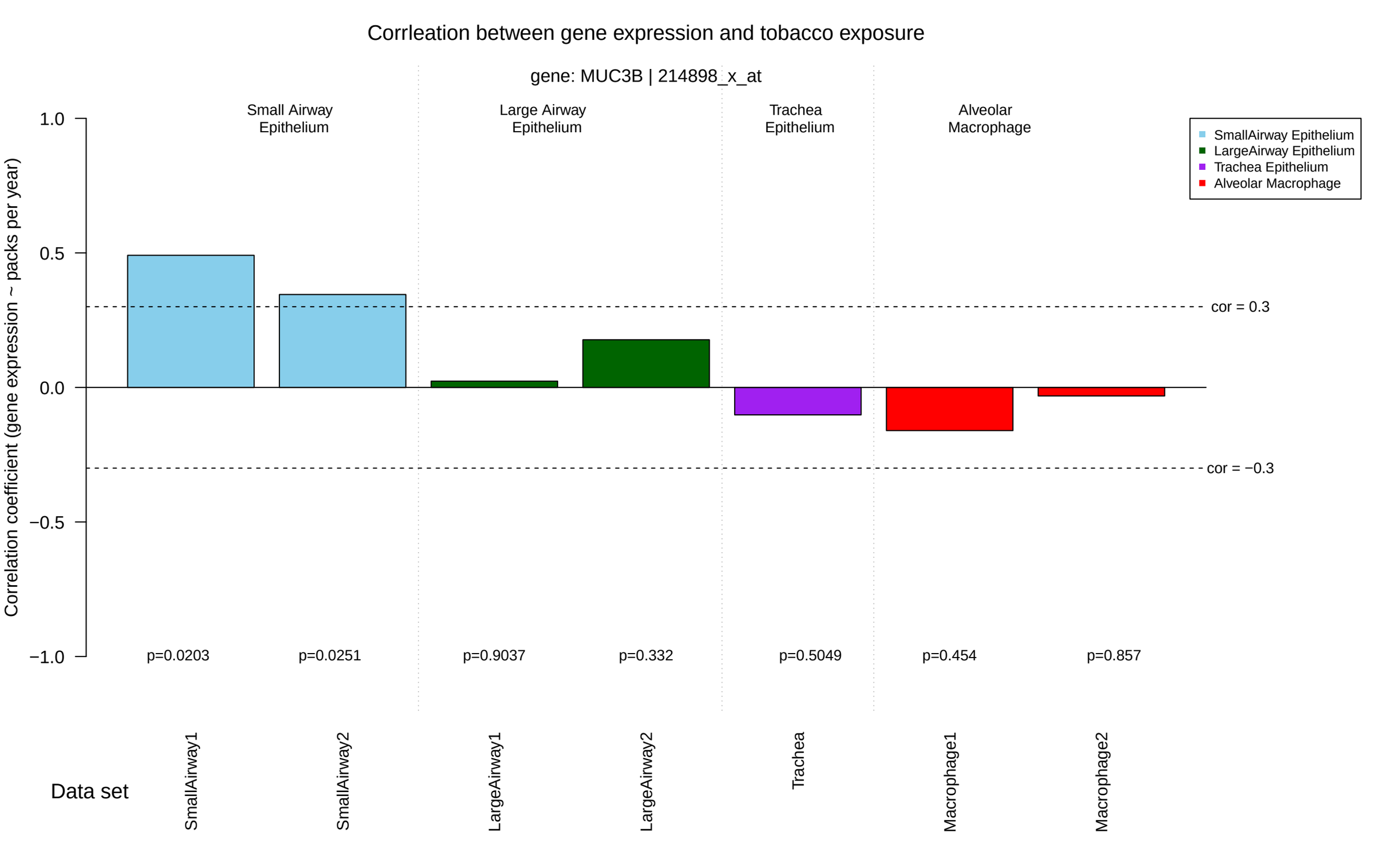

set: LargeAirway1 (p=0.1318)
MUC3B | 214898_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0261)
MUC3B | 214898_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6158)
MUC3B | 214898_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9352)
MUC3B | 214898_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3764)
MUC3B | 214898_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MUC3B | 214898_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0203
p=0.0251
p=0.9037
p=0.332
p=0.5049
p=0.454
p=0.857
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2